

Slip-Cast Ceramic Udu Drum Family — v4.1 Capstone

Instrument-Maker Print Packet

Build packet folder: /sessions/friendly-stoic-shannon/mnt/GitHub/udu

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This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

File Map

File	Purpose
design.md	Project intent, catalog metadata, assumptions, and validation plan.
bom.csv	Starter bill of materials with part categories, quantities, drawing refs, and notes.
sourcing.csv	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
cut-list.csv	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
drawing-brief.md	Manufacturing drawing and technical product sketch brief.
assembly-manual.md	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
validation.csv	Target/measured values, tolerance, environment, result, and tuning/build action log.
supplier-rfq.md	Supplier email/request-for-quote starter.
visual-bom-brief.md	Art direction for an image-forward visual BOM.
wolfram-starter.wl	Wolfram starter for physics, optimization, visualization, and validation.
README.md	Project artifact.
family-spec.csv	Project artifact.
risks.md	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Slip-Cast Ceramic Udu Drum Family

Source

- Design table: `udu-design-table.xlsx` (Master_Inputs + Design_Table sheets)
- Family spec: `family-spec.csv` (S/M/L/XL members)
- SW reference: `sw-reference/` (MasterLayout convention, equation list, design-table import procedure)
- Workbook lineage: `docs/Udu.xlsx`, sheet `Udu Drum Family`, inspected range `A1:H146`
- Workbook content observed: design inputs, dual-Helmholtz calculator, four-size family, variants, playing techniques, ceramic production workflow, BOM, design notes, Wolfram notebook notes.

Project Intent

Build a slip-cast ceramic udu family — Small / Medium / Large / XL — with repeatable chamber volumes and two playable Helmholtz tones: the main mouth tone and the side-hole slap tone. The first production target is the **Medium** standard udu (UDU-P2: 10 in × 12 in body, B3 mouth / G3 side), followed by a scaled family. The udu (Igbo: **ndi**, "vessel") is a Nigerian clay-vessel drum and the documentation here treats that lineage as fact, not marketing.

The repository is the engineering artifact for the dual-Helmholtz coupled-resonator model: parametric design table → 3D-printed master halves → two-piece plaster mold → Cone-6 stoneware slip-cast → bisque + tune + glaze. The same pipeline serves `ocarina/`, `gemshorn/`, and `transverse-flute/`.

Governing Model

The udu uses a shared chamber with two coupled Helmholtz resonators:

$$f_{\text{top}} = c / (2\pi) * \sqrt{A_{\text{top}} / (V * L_{\text{top}})}$$

$$f_{\text{side}} = c / (2\pi) * \sqrt{A_{\text{side}} / (V * L_{\text{side}})}$$

$$L = \text{wall_thickness} + 0.6 * \sqrt{A/\pi}$$

$$V = (\pi/6) * \text{body_diam}^2 * \text{body_height} * \text{shape_factor}$$

When both ports are open and the player's hands move, the modes couple — the single-port formulas are useful first-order targets, and the placeholder coupled-mode eigenvalue model lives in ``wolfram-starter.wl``. Measured prototypes define the final tuning corrections; per-family corrections feed back through ``scripts/record_measurement.py`` (see ``references/empirical-learning-loop.md``).

Hardware Alignment

This repo is one of the slip-cast targets for the in-flight Bambu printer + ceramic kiln pipeline shared with ``ocarina/``, ``gemshorn/``, and ``transverse-flute/``. The build chain:

Stage	Tool / pipeline	Notes
Design table	<code>`udu-design-table.xlsx`</code>	Master_Inputs Blue cells = inputs; volume formula derived
Master scaling	<code>`master_scale_factor = 1/(1 - measured_shrinkage)`</code>	Re-derive per slip batch — see risks.md SUP-01
Master print	Bambu X1C (UDU-S, UDU-M) or outsourced print (UDU-L, UDU-XL)	X1C envelope ≈ 256 mm; L/XL exceed it — see risks.md SUP-02
Mold	Two-piece plaster mother mold, USG #1 pottery plaster	Split line off the hand-contact equator
Cast	Cone 6 stoneware casting slip	Shrinkage coupon required per batch
Trim	Side hole cut at leather-hard, undersized	Final tune at bisque
Fire	Cone 06 bisque $\rightarrow$ Cone 6 glaze	Glaze exterior only
Validate	<code>`validation.csv`</code> mouth/side/coupled measurements	<code>`record_measurement.py`</code> updates per-family corrections
CAD round-trip	<code>`UDU-000_MasterLayout.SLDPRT`</code> $\leftrightarrow$ <code>`cad/sw-design-table.xlsx`</code>	See <code>`sw-reference/`</code> for the SW workflow

Current Workbook Inputs

Field	Current value
Body diameter	10 in
Body height	12 in
Wall thickness	0.3 in
Mouth diameter	3 in
Side hole diameter	2 in

Shape factor	0.7
Clay body	Cone 6 stoneware
Shrinkage	0.12
Speed of sound	13510 in/s

Current Family Targets

Model	Diameter	Height	Volume	Mouth tone	Side tone	Use
Small	8 in	10 in	235 in³	D#4	A3	Tabletop and high tones
Medium	10 in	12 in	440 in³	B3	G3	Most versatile standard size
Large	12 in	14 in	739 in³	G#3	D3	Deep bass and traditional feel
XL	14 in	16 in	1149 in³	F3	C3	Concert bass, likely needs stand

Critical Design Features

- Body volume controls both main tones; slip casting is valuable because it makes volume repeatable.
- Port diameters and effective neck lengths are the main pitch controls.
- Wall thickness controls shell tap tone, durability, drying risk, and weight more than Helmholtz pitch.
- The side hole edge must be comfortable for repeated hand slaps.
- The mouth rim should be rounded enough for comfort but consistent enough for repeatable hand sealing.
- Feet, stand interface, or flat area should be designed before the mold if the instrument needs stability.

Cultural And Product Note

The udu ha

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	category	item	qty	spec	make_buy	estimated_cost	source_note	drawing_ref	notes
UDU-BOM-001	Clay	Cone 6 casting slip	5 gal	Commercial pre-mixed stoneware	Buy	\$25-50	Workbook estimate	UDU-DRAW-001	Enough for roughly 8-12 medium
UDU-BOM-002	Mold	#1 pottery plaster	50 lb	Pottery plaster for 2-piece mold	Buy	\$25-35	Workbook estimate	UDU-DRAW-002	Record plaster ratio and mold
UDU-BOM-003	Master	3D printed master	2 halves	PLA master scaled for shrinkage	Make	\$5-15	Workbook estimate	UDU-DRAW-003	Sand and seal before mold making
UDU-BOM-004	Mold	Mold soap/release	1 bottle	Plaster-compatible	Buy	\$8-12	Workbook estimate	UDU-DRAW-004	Use on sealed master and plaster
UDU-BOM-005	Cutting	Hole cutter set	1 set	1.5 in to 3 in round cutters	Buy	\$10-20	Workbook estimate	UDU-DRAW-005	Cut side hole at leather-hard stage
UDU-BOM-006	Firing	Bisque and glaze	2 firings	Cone 06 bisque and Cone 6 glaze	Buy	\$20-50/fire	Workbook estimate	UDU-DRAW-006	Record schedule and kiln.
UDU-BOM-007	Finish	Exterior glaze	1 pint	Cone 6 matte or satin glaze	Buy	\$10-20	Workbook estimate	UDU-DRAW-007	Leave interior unglazed.
UDU-BOM-008	Hardware	Rubber feet	1 set	Three low-profile feet	Buy	\$3-5	Workbook estimate	UDU-DRAW-008	Optional; test effect on shell resonance
UDU-BOM-009	Electronics	Contact pickup	1	Piezo disc/contact mic	Buy	\$15-30	Workbook estimate	UDU-DRAW-009	Optional electric variant.
UDU-BOM-010	Measurement	Water-fill and frequency measure	1	Calibrated vessel and tuner/mic	Buy	TBD	Add supplier/date	UDU-DRAW-010	Required for empirical correction

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item_id	item	required_specs	search_terms	supplier_candidates	date_checked	unit_price	lead_time	substitution_risk	note
UDU-SRC-001	Cone 6 casting slip	stoneware casting slip	slip 6 stoneware	5 suppliers	15 days	100 lbs bag	1 week	Substitute only with known glaze	Unshrinkable, no changes v
UDU-SRC-002	#1 pottery plaster	Absorbent pottery slip	#1 plastering	5 suppliers	5 days	50 lbs bag	1 week	Equivalent pottery plaster	Must plaster before is about 10
UDU-SRC-003	PLA filament	Stable print material	PLA filament	10 suppliers	10 days	100 g spool	1 week	Resin or outsource printing	Can print any shape, sealed v
UDU-SRC-004	Mold release	Plaster-compatible	pottery mold release	5 suppliers	5 days	100 g jar	1 week	Use known ceramic release	Release failure can destroy the
UDU-SRC-005	Hole cutters	Clean round cutters	ceramic hole cutters	5 suppliers	10 days	100 g jar	1 week	Custom brass tool	Brass tool can hurt p
UDU-SRC-006	Rubber feet	Low profile feet	rubber feet	5 suppliers	5 days	100 g jar	1 week	Use removable feet	Feet can drop, lose tone.
UDU-SRC-007	Piezo pickup	Contact pickup	piezo pickup	5 suppliers	5 days	100 g jar	1 week	External contact	Interchangeable and prototype

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

cut_id	part	qty	rough_dim	final_dim	material	orientation	operation	tolerance_in	yield_or_offcut	notes
UDU-CUT-001	Master body upper half (3D-printed)	12	12 x 12 x 12	Medium master scaled	FLY (0.5 grain)	Random	Print 130 x 130 x 130	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-002	Master body lower half (3D-printed)	12	12 x 12 x 12	Medium lower of UDU-001	FLY (0.5 grain)	Random	Print 130 x 130 x 130	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-003	Wooden cottle boards (mold making)	16	16 x 0.75 x 14	14 x 0.75	Melamine-faced	Smooth	Hand cut	0.005	100%	Base sheet of 4' x 8' x 1/8" plywood; use for 16 boards
UDU-CUT-004	Plaster mother mold half (pouring)	1	16 x 14 x 3.5	16 x 14 x 3.5	US Gypsum	Random	Hand cut	0.005	100%	Use for 16 boards; if low, discard; possible to re-use for other parts
UDU-CUT-005	Greenware vessel (slip-cast)	1	10 x 12 x 12	10 x 12 x 12	Greenware	Random	Slip-cast	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-006	Mouth opening (cast as part of master)	1	3.0 in dia	3.0 in dia	FIRE	Random	Cast	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-007	Side hole (leather-hard)	1	2.0 in dia	2.0 in dia	FIRE	Random	Cast	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-008	Foot pads (optional)	1	0.5 in dia	0.5 in dia	FIRE	Random	Cast	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-009	Post-bisque tuning (grind)	1	Within +/- 50	Within +/- 50	Post-bisque	Random	Grind	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-010	Wax resist mask (application)	1	~0.020 in dry	~0.020 in dry	Wax	Random	Apply	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-011	Glaze application (optional)	1	~0.010-0.020 in	~0.010-0.020 in	Glaze	Random	Apply	0.005	100%	Re-use; if low, discard; possible to re-use for other parts
UDU-CUT-012	Family scaling	1	Small/Large	Small/Large	Family scaling	Random	Scale	0.005	100%	Re-use; if low, discard; possible to re-use for other parts

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Udu Drum Family Drawing Brief

Required Views

- Front view with overall height, max body diameter, mouth diameter, and side-hole position.
- Side view showing body profile, wall target, side-hole axis, and foot/stand contact.
- Top view showing mouth opening, rim radius, and symmetry datums.
- Section view through mouth and side hole showing wall thickness and effective neck length.
- Mold split view with registration keys, plaster thickness, pour/drain direction, and seam-cleanup zone.
- Optional electric variant view showing piezo location, wire path, jack location, and service access.

Critical Dimensions

Dimension Source Tolerance intent
--- --- ---
Chamber volume Workbook formula and water-fill measurement Tuning critical
Body diameter and height Workbook inputs Tuning and ergonomics
Mouth diameter Workbook input Tuning and hand seal
Side-hole diameter Workbook input and tuning log Tuning and playability
Wall thickness Workbook input and cast measurement Tap tone/durability
Rim radii CAD and ergonomic test Comfort
Master scale factor Measured clay shrinkage Process critical

Notes For CAD

- Use a named shape factor or profile parameter so S/M/L/XL can be regenerated from the same model.
- Keep side-hole position as a parameter tied to hand comfort, not only aesthetics.
- Design any feet as optional variants so acoustic damping can be tested.
- Keep the water-udu variant separate because water fill and leak-tightness change requirements.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Udu Drum Family Assembly Manual

Scope

This manual covers slip-cast ceramic udu prototypes based on `docs/Udu.xlsx`, starting with the medium standard udu and expanding to a scaled family.

Tools

- CAD package for body profile and mold split.
- Large-format 3D printer or outsourced print for master halves.
- Plaster mold-making setup: cottle boards, scale, bucket, plaster, mold soap, clamps.
- Cone 6 stoneware casting slip.
- Hole cutters, ribs, sponges, fettling knife, rasp, and small files.
- Calipers, wall-thickness gauge if available, graduated water-fill vessel, tuner/microphone.
- Kiln access, wax resist, exterior glaze.
- Optional: rubber feet, stand prototype, piezo/contact mic.

Process

1. ****Choose size****
 - Start with the medium workbook design unless testing mold workflow at smaller scale.
 - Assign build ID and expected shrinkage.
2. ****CAD body and mold strategy****
 - Model the target fired body first.
 - Add shrinkage scale to produce the master body.
 - Choose a split line that releases cleanly and avoids hand-contact ridges.
3. ****Print and seal master****
 - Print master halves rigid enough for plaster.
 - Sand high layer lines, fill defects, and seal.
 - Verify master diameter and height before molding.
4. ****Make plaster mold****

- Apply release.
- Pour mold halves with registration keys.
- Dry completely before first casting.
- Record mold dry weight and first-use date.

5. **Slip cast**

- Fill mold with slip.
- Drain after target wall thickness buildup.
- Record fill time, drain time, slip batch, room condition, and demold time.

6. **Leather-hard work**

- Clean seam.
- Cut side hole undersized or at first target.
- Round mouth and side-hole edges.
- Add foot/stand features only if they do not trap drying stress.

7. **Dry**

- Dry under plastic for even moisture.
- Rotate periodically if needed.
- Inspect rim, side hole, foot, and seam.

8. **Bisque fire**

- Fire to the clay body's bisque schedule.
- Measure dimensions, chamber volume, mass, and pitch.

9. **Tune**

- Tune with small changes and document each pass.
- For simple open-port Helmholtz behavior, larger port area usually raises resonance; hand-covered playing conditions should be measured separately.
- Do not remove more material until the measurement setup is repeatable.

10. **Glaze**

- Glaze exterior only unless intentionally testing interior glaze damping.
- Keep mouth and side-hole edges clean.

11. **Final validation**

- Measure final tones, shell tap, weight, stability, and ergonomics.
- Record whether feet or stands damp the ceramic tap tone.

Failure Modes To Watch

- Cracks at side hole: hole cut too late, edge too sharp, or drying too fast.
- Dead bass response: chamber too small, ports poorly proportioned, walls too thick, or hand seal poor.
- Uncomfortable slap edge: rim needs larger radius or smoother fired edge.
- Warped base: unsupported drying or uneven mold thickness.
- Over-damped shell: glaze too heavy, feet too soft, or wall too thick.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

build	stage	date	model	clay	body	ink	age	steps	pedestal	lip	height	rib	band	width	center	tilt	in	dia	target	target	freq	height	freq	action	result	notes
UDU-P1	greenware		Small					8	10	TBD	TBD	3	2	mouth	open				311							First slip-cast body ch
UDU-P1	bisque		Small					TBD	TBD	TBD	TBD	TBD	TBD	mouth	open				311							Measure after shrinka
UDU-P1	bisque		Small					TBD	TBD	TBD	TBD	TBD	TBD	side	open	43			223							Check hand slap resp
UDU-P2	greenware		Medium					10	12	0.3	440	3	2	mouth	open				249							Workbook standard ta
UDU-P2	bisque		Medium					TBD	TBD	TBD	TBD	TBD	TBD	mouth	open				249							Primary tuning checky
UDU-P2	bisque		Medium					TBD	TBD	TBD	TBD	TBD	TBD	side	open	38			192							Record whether perce
UDU-P2	glaze_fire		Medium					TBD	TBD	TBD	TBD	TBD	TBD	mouth	open				249							Measure glaze shift.
UDU-P3	bisque		Water udu					TBD	TBD	TBD	TBD	TBD	TBD	tilt	sweet	TBD			TBD							Record water amount

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Udu Drum Family (UDU-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

****Subject:**** RFQ — slip-cast ceramic udu drum prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic udu drums (Igbo-rooted clay vessel drum, S/M/L/XL sizes, ~8-14 in body diameter, ~0.30 in / 7.6 mm wall) and need quotes for the following. The first build is a Medium standard udu (10 in dia × 12 in tall, target mouth tone B3 ≈ 247 Hz, side-hole tone G3 ≈ 196 Hz) based on the parametric dual-Helmholtz design table at [tonykoop/udu](https://github.com/tonykoop/udu).

Items

| # | Item | Required spec | Approx. qty (per prototype run) |

|---|---|---|---|

| 1 | Cone 6 stoneware casting slip | Mature Cone 6 stoneware slip with ****published shrinkage**** (X/Y/Z), ****water absorption %****, and ****firing schedule****. Larger pour volumes than typical small-vessel work — slip needs to flow predictably into a 12+ in master cavity without cracking on drain. | 5 gal (Medium) — 15 gal (S/M/L/XL set) |

| 2 | #1 pottery plaster (large mold scale) | USG #1 Pottery Plaster (or equivalent) suitable for a 16 × 14 × 4 in two-piece pour. Need ****water:plaster ratio****, ****set time****, and ****dimensional stability**** (creep) over months of use. | 50 lb (single Medium mold) — 200 lb (full S/M/L/XL set) |

| 3 | Mold release / mold soap | Plaster-compatible parting agent. Larger surface area than small vessels — needs to release cleanly off a 12+ in body without locking. | 2 bottles / 32 oz |

| 4 | Cone 6 exterior glaze | Satin or matte glaze. ****Color: TBD**** — earth tones / iron oxide / rust red are visually closer to traditional fired Igbo-clay ware. Glaze must NOT bridge mouth or side-hole edges (acoustic-critical). | 1-2 pints per body |

| 5 | Brushable wax resist | Standard ceramic wax resist for masking mouth rim, side-hole rim, and chamber interior pre-glaze. | 1 jar / 16 oz |

| 6 | Hole cutter set | Round cutters, ****1.5 in to 3 in diameter****, for greenware side-hole cutting. Brass tube cutters or commercial hole-cutter discs both acceptable. | 1 set |

| 7 | Diamond needle file / burr set (post-bisque tuning) | Assorted profiles, ****3-10 mm****; resin-bonded acceptable. Used to enlarge ports for pitch correction. | 1 set |

| 8 | Bisque + glaze firing service (if local) | Cone 06 bisque + Cone 6 glaze fire; medium-large kiln (10 ft³+) can fit XL family. | 2 fires per body |

| 9 | (Optional) Master printing / outsource | Large-format PLA or resin print, up to **~14 in master width** with **0.010 in dimensional accuracy**. We currently print in-house but want a backup option. | 1-4 master pairs (S/M/L/XL) |

| 10 | (Optional) Rubber feet — low-profile | Self-adhesive low-profile rubber feet, ~0.5 in dia, for floor stability. Test-fit only — feet damp shell tap tone, so this is an acoustic trade-off. | 3 per body |

| 11 | (Optional) Piezo / contact pickup | Contact pickup that can survive being epoxied to interior wall OR magnetically/clip-mounted externally. | 1 per electric variant |

What we need in your quote

- **Unit price** and any volume breaks
- **Minimum order quantity**
- **Current stock status** and **lead time**
- **Shipping estimate** to 94566 (Pleasanton, CA) — or local pickup option
- **Material safety data sheet (MSDS)** and **technical data sheet** for items 1, 2, 4, 5
- **Recommended substitutions** if anything goes out of stock
- For item 1: most recent **shrinkage test data** for your batch (we re-measure on our side, but a baseline matters)
- For item 9: **photos of similar-sized prints** showing layer-line consistency and dimensional check process

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, fired chamber volume (water-fill), wall-thickness consistency at rim/shoulder/belly/foot, and acoustic tuning at TWO conditions: mouth-tone (mouth open, side covered) and side-hole tone (side open, mouth covered or hand-muted). Target tuning tolerance: **±50 cents post-bisque** on both modes for the Medium body. Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $1/(1-0.12) = 1.136$), so reproducibility data matters more than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Cultural note for sales / marketing teams

The udu has Igbo (Nigerian) roots and is not a generic ceramic bass pot. We're not asking for marketing positioning — we're asking for materials. But please don't include "decorative African pot" or similar reductive copy in any spec sheet response. Lineage attribution belongs to the player and the documentation, not the supplier.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/udu (private; access on request)

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Udu Visual BOM Brief

Goal

Create a one-page visual BOM for the slip-cast udu family that shows the body, mold, ports, finishing choices, and optional electric/water variants.

Layout

- Header: "Slip-Cast Ceramic Udu Drum Family".
- Hero image: medium udu render or finished prototype.
- Family strip: small, medium, large, XL silhouettes with target pitch ranges.
- Exploded/process view: 3D printed master halves, plaster mold halves, cast body, side-hole cut, fired body.
- BOM table: item number, material/tool, quantity, spec, cost estimate, make/buy, image.
- Physics inset: top-port and side-port Helmholtz formulas.
- Measurement inset: body volume, mouth tone, side tone, wall thickness, mass.

Image Requirements

Use real photos as soon as available:

- Printed master.
- Plaster mold halves.
- Leather-hard cast before side-hole cutting.
- Side-hole close-up.
- Fired body with hand scale.
- Optional pickup or rubber-foot detail.

Generated or CAD images should be marked as placeholders until replaced by shop photos.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

(* Udu drum family Helmholtz notebook starter *)

```
ClearAll["Global`*"];
```

```
c = 13510; (* in/s at about 68 F *)
```

```
areaCircle[d_] := Pi*(d/2)^2;
```

```
leff[wall_, area_] := wall + 0.6*Sqrt[area/Pi];
```

```
volumeOvoid[diam_, height_, shapeFactor_] := Pi/6*diam^2*height*shapeFactor;
```

```
helmholtzHz[area_, volume_, neckLength_] :=
```

```
(c/(2*Pi))*Sqrt[area/(volume*neckLength)];
```

```
centsError[measured_, target_] := 1200*Log[2, measured/target];
```

```
uduModel[name_, diam_, height_, wall_, mouthDiam_, sideDiam_, shapeFactor_] :=
```

```
Module[{v, aMouth, aSide, lMouth, lSide},
```

```
v = volumeOvoid[diam, height, shapeFactor];
```

```
aMouth = areaCircle[mouthDiam];
```

```
aSide = areaCircle[sideDiam];
```

```
lMouth = leff[wall, aMouth];
```

```
lSide = leff[wall, aSide];
```

```
<|
```

```
"Name" -> name,
```

```
"VolumeIn3" -> v,
```

```
"MouthHz" -> helmholtzHz[aMouth, v, lMouth],
```

```
"SideHz" -> helmholtzHz[aSide, v, lSide],
```

```
"IntervalSemitones" -> 12*Log[2, helmholtzHz[aMouth, v, lMouth]/helmholtzHz[aSide, v, lSide]]
```

```
|>
```

```
];
```

```
models = {
```

```
uduModel["Small", 8, 10, 0.3, 3, 2, 0.7],
```

```
uduModel["Medium", 10, 12, 0.3, 3, 2, 0.7],
```

```
uduModel["Large", 12, 14, 0.3, 3, 2, 0.7],
```

```
uduModel["XL", 14, 16, 0.3, 3, 2, 0.7]
```

```
};
```

Dataset[models]

(* First coupled-mode placeholder: replace kCouple with measured fit. *)

coupledFrequencies[f1_, f2_, kCouple_] :=

Sqrt[Eigenvalues[{{f1^2, kCouple}, {kCouple, f2^2}}]];

README.md

Project artifact.

Udu — Slip-Cast Ceramic Vessel Drum Family

> *Engineering documentation for a 3D-printed-master, plaster-mold, cone-6-stoneware slip-cast udu drum family — from dual-Helmholtz physics through the parametric design table to a manufacturable build packet.*

![[Hero — AI-generated concept render of a Medium slip-cast udu (placeholder until first prototype is photographed)]](images/hero-concept.png)

AI-generated concept render — replace with photo of first UDU-P2 (Medium) prototype after firing. Manufacturing dimensions come from the parametric design table, not from this image.

What this is

Engineering documentation for a four-size family (Small / Medium / Large / XL) of slip-cast ceramic udu drums, built around the **dual-Helmholtz coupled-resonator** model and a 3D-printed-master / plaster-mold / Cone-6-stoneware production pipeline. The repository combines:

1. **A parametric design table** ([`udu-design-table.xlsx`](udu-design-table.xlsx)) — body diameter, body height, wall thickness, mouth diameter, side-hole diameter, shape factor, shrinkage, and a four-size family-spec block with target mouth-tone and side-tone frequencies. Master scale factor $\frac{1}{(1 - \text{measured_shrinkage})}$ is parameterized; for a 12% shrinkage assumption, the master scales at $\sim 1.136\times$.
2. **A full build packet** ([`design.md`](design.md), [`bom.csv`](bom.csv), [`sourcing.csv`](sourcing.csv), [`cut-list.csv`](cut-list.csv), [`validation.csv`](validation.csv), [`assembly-manual.md`](assembly-manual.md), [`supplier-rfq.md`](supplier-rfq.md), [`drawing-brief.md`](drawing-brief.md), [`visual-bom-brief.md`](visual-bom-brief.md), [`wolfram-starter.wl`](wolfram-starter.wl)) — the same scaffold used across the [`tonykoop`](https://github.com/tonykoop) musical-instrument catalogue.
3. **A capstone slide deck and printable shop packet** ([`capstone-deck.pptx`](capstone-deck.pptx), [`print-packet.pdf`](print-packet.pdf)) — recruiter-facing artifacts that show the design is documented well enough that someone else could build it.

Sister repos: [`gemshorn`](https://github.com/tonykoop/gemshorn) (slip-cast horn-flute mold workflow), [`ocarina`](https://github.com/tonykoop/ocarina) (single-Helmholtz vessel-flute, sister slip-cast target), [`transverse-flute`](https://github.com/tonykoop/transverse-flute) (slip-cast workflow at larger scale), [`djembe`](https://github.com/tonykoop/djembe) (where Helmholtz-cavity-resonator analysis was originally derived for the bass tone), and [`instrument-maker`](https://github.com/tonykoop/instrument-maker) (the agent skill that generated this packet).

Background — what makes the udu different

The udu (Igbo: **■d■**, "vessel") is a Nigerian clay-vessel drum: a closed ceramic body with a mouth at the top and a single side hole. Striking the mouth with a flat hand produces one resonant tone; striking the side hole produces another; opening, closing, and partly hand-covering each port lets the player slide and sweep between the two pitches. **It is not a generic ceramic bass pot.** Its lineage is specific and the documentation here treats that lineage

as fact, not as marketing copy.

The governing model is **two coupled Helmholtz resonators sharing a chamber volume**:

$$f_{\text{top}} = c / (2\pi) \cdot \sqrt{A_{\text{top}} / (V \cdot L_{\text{top}})}$$

$$f_{\text{side}} = c / (2\pi) \cdot \sqrt{A_{\text{side}} / (V \cdot L_{\text{side}})}$$

with $L = \text{wall_thickness} + 0.6 \cdot \sqrt{A/\pi}$ (flanged-port end correction) and a coupling term that becomes important when both ports are partly open. Single-port formulas are useful **first-order** targets; the coupled-mode behavior — and especially the perceived pitch when a player's hand partly covers either port — has to be measured empirically. See design.md and wolfram-starter.wl for the full treatment, including a placeholder eigenvalue model for coupled frequencies.

The practical consequence is that udu tuning is **two parametric pitches plus a player-coupled middle range**. Body volume sets the bass-end ceiling; port diameters set the upper bound; wall thickness and shape factor are secondary. Hand coupling is a sweep, not a discrete control — which is why the design table targets fall within ± 50 ¢ rather than the ± 25 ¢ used for fixed-pitch instruments like the ocarina.

Family targets

The first prototype is the **Medium** body (10 in dia × 12 in tall, target mouth tone B3 ≈ 247 Hz, side tone G3 ≈ 196 Hz, chamber volume ≈ 440 in³). Once Medium voicing is stable, the same parametric model drives Small / Large / XL — see design.md and the family-spec block of udu-design-table.xlsx.

Model	Body diameter	Body height	Volume	Mouth tone	Side tone	Use
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Small	8 in	10 in	~235 in³	D $\blacksquare$ 4	A3	Tabletop, high tones
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Medium	(prototype 1)	10 in	12 in	~440 in³	B3 G3	Most versatile standard size
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Large	12 in	14 in	~739 in³	G $\blacksquare$ 3	D3	Deep bass, traditional feel
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XL	14 in	16 in	~1149 in³	F3 C3		Concert bass; needs stand
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Prototype ladder:

Prototype	Goal	Success criteria
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UDU-P0	cup/port tile	Validate clay wall + tap behavior No cracks; clear ceramic tap; clean port edge
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UDU-P1	Small body	Validate slip casting + side-hole slap Measurable Helmholtz tones, comfortable slap edge
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UDU-P2	Medium body	Hit workbook Medium target Mouth and side tones within ± 50 ¢ post-bisque
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UDU-P3	Water udu	Pitch-bending volume variant Smooth tilt-bend without leaking
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UDU-P4	Family molds	Scale S/M/L/XL Predictable pitch trend across sizes
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Hardware alignment — Bambu printer + ceramic kiln

This repo is one of the ****slip-casting targets**** for the in-flight Bambu printer + kiln pipeline. The build chain is:

1. Parametric design table sets fired-body dimensions.
2. Master scale factor $\frac{1}{(1 - \text{measured_shrinkage})}$ scales the master STL.
3. Bambu print master halves in PLA (or sealable resin); sand layer lines and seal. *Note: the Medium body is right at the edge of the Bambu X1C build envelope; XL will need a print farm split or outsourced master.*
4. Two-piece plaster mother mold around the sealed master.
5. Slip-cast Cone 6 stoneware → demold leather-hard → cut side hole undersized → bisque → tune → glaze (exterior only) → final fire.
6. Validate against `[validation.csv]`(validation.csv); fold corrections back into the design table.

The same pipeline serves the `[ocarina]`(<https://github.com/tonykoop/ocarina>), `[gemshorn]`(<https://github.com/tonykoop/gemshorn>), and `[transverse-flute]`(<https://github.com/tonykoop/transverse-flute>) repos.

Repository structure

udu/

■■■ README.md ← you are here

■■■ LICENSE ← CC-BY 4.0

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■■■ design.md ← dual-Helmholtz model, design intent, prototype ladder

■■■ udu-design-table.xlsx ← parametric spreadsheet (formulas, blue inputs, family block)

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■■■ bom.csv ← bill of materials

■■■ sourcing.csv ← supplier/search tracker

■■■ cut-list.csv ← rough/finished dimensions, tolerances, family scaling

■■■ validation.csv ← target/measured tuning + cents-error log

■■■ supplier-rfq.md ← RFQ template for slip / plaster / consumables

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■■■ assembly-manual.md ← shop-floor build sequence

■■■ drawing-brief.md ← required views + critical dimensions for CAD

■■■ visual-bom-brief.md ← visual-BOM art-direction brief

■■■ wolfram-starter.wl ← dual-Helmholtz physics starter notebook

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■■■ capstone-deck.{md,pptx} ← capstone slide deck (9 slides — v1 codex skill output)

■■■ print-packet.{md,html,pdf} ← combined print-ready shop packet

■■■ capstone-manifest.json ← orientation manifest



- cad/ ← parametric body OpenSCAD starter (family-aware)
- cnc/ ← (deferred — slip-cast does not need CNC unless turning a wooden master)
- drawings/ ← SVG drawings: section, mold split, family scale
- images/ ← AI-generated concept renders (placeholders)

Status

| Section | Status |

|---|---|

- | Parametric design table + dual-Helmholtz model | ✓ done |
- | Build packet (BOM / sourcing / cut-list / validation / RFQ) | ✓ done |
- | Assembly manual + drawing brief | ✓ done |
- | Wolfram physics starter (with coupled-mode placeholder) | ✓ done |
- | Capstone deck + print packet | ✓ done (auto-generated, recruiter-facing) |
- | Concept renders (AI-generated, captioned) | ✓ done (placeholders) |
- | Parametric CAD (OpenSCAD starter, family-aware) | ✓ done (master-shape only) |
- | Dimensioned drawings (SVG) | ✓ done (section + mold split + family scale) |
- | Production-ready CAD (.step / .stl) | **deferred** — generated after empirical Medium-body validation |
- | Coupled-mode tuning model (eigenvalue fit) | **deferred** — needs Phase-1 measured data |
- | First UDU-P1 (Small) prototype build | forthcoming (Bambu + kiln pipeline) |

Tier 3 production files (.step, validated .stl, .dxf, .gcode) are **out of scope** until UDU-P2 (Medium) validates wall thickness and port-pitch behavior. See design.md "Open Assumptions" for the deferral reasoning.

License

Released under [CC-BY 4.0](LICENSE) — original written content, design files, photographs, and physics work in this repository are mine, free to reuse and adapt with credit.

The **udu** as an instrument concept and name belongs to the Igbo people of Nigeria; the lineage attribution in design.md is part of the documentation, not a license claim.

family-spec.csv

Project artifact.

	member	target	target	length	diameter	density	height	width	volume	shape	material	color	weight	temperature	humidity	location	status	notes	
UDU-S	D#4	311.13	A3	220.00	8.0	10.0	0.30	3.0	2.0	0.70	234.6	340.8	262.4	+157	+304	1.136	Cone 6	new	Small/tabletop. Predict
UDU-M	B3	246.94	G3	196.00	10.0	12.0	0.30	3.0	2.0	0.70	439.8	248.9	191.6	+14	-39	1.136	Cone 6	new	Medium/standard. Wo
UDU-L	G#3	207.65	D3	146.83	12.0	14.0	0.30	3.0	2.0	0.70	738.8	192.1	148.0	-126	+14	1.136	Cone 6	new	Large/deep bass. Mo
UDU-XL	F3	174.61	C3	130.81	14.0	16.0	0.30	3.0	2.0	0.70	1148.7	153.9	118.5	-220	-172	1.136	Cone 6	new	Concert bass. Predict

risks.md

Project artifact.

Udu Drum Family — Risk Register

- > Red-team pass on the slip-cast ceramic udu family (UDU-S/M/L/XL). Every risk
- > below has a verifiable test attached. High-severity risks are flagged for
- > Tony's decision before we commit material.

This document is the v4 obligation produced by the `red-team` specialist; the verifier asks "is this packet internally consistent?" while red-team asks "is this design actually safe to build?" The two are independent — a polished self-consistent packet can still fail at the bench.

Categories enumerated below: **Acoustic**, **Structural**, **Ergonomic**, **Supply**, **Fit/Finish**.

Acoustic

[Acoustic] Predicted frequencies diverge from workbook targets at family extremes — HIGH

Symptom: UDU-S mouth tone reads ~340 Hz vs D $\sharp$ 4 target 311 Hz (+157 ¢ sharp); UDU-XL mouth tone reads ~154 Hz vs F3 target 175 Hz (-220 ¢ flat). Family targets at the extremes are out of reach of bisque-stage trim alone (the trim band is roughly ± 100 ¢ before the port becomes structurally compromised or visually disproportionate).

Mechanism: The workbook used identical port diameters (mouth 3 in, side 2 in) and a fixed shape factor 0.70 across the family. Helmholtz frequency scales as $\sqrt{A / (V \cdot L)}$, so holding A constant while V scales linearly makes the predicted pitch scale as $V^{-1/2}$. The targets were chosen on a musical interval ladder (mouth tones D $\sharp$ 4 / B3 / G $\sharp$ 3 / F3 — descending major 3rd / minor 3rd / minor 3rd) but the *acoustic* scaling of the chosen geometry doesn't match the *musical* scaling of the targets.

Test: Cast a one-off small body (UDU-P1) and measure the mouth tone with a tuner before the family molds are committed. Cents error >100

on the smallest member confirms the family-level mismatch.

****Mitigation:****

- Drop the UDU-S mouth diameter to ≈ 2.5 in (shrinks predicted Hz back toward 310 Hz) and grow the UDU-XL mouth toward ≈ 3.5 – 4.0 in (grows predicted Hz back toward 175 Hz).
- *Or* keep ports constant and shift the family targets onto an acoustically-honest ladder (e.g., mouth tones at fixed volume ratios).
- Document whichever path is chosen in `family-spec.csv` and re-emit the per-member SVG drawings.

****Severity:**** ****High**** — design change required, not just a manufacturing adjustment. Escalated to Tony for decision before Tier-3 production CAD.

[Acoustic] Coupled-mode pitch when both ports are partly open is not predicted — Medium

****Symptom:**** A real udu's musical character is the *sweep* between the mouth and side tones as the player covers/uncovers each port. Single-port formulas (used in the design table and Wolfram starter) do not predict the coupled-mode pitch or the transition curve.

****Mechanism:**** Two ports sharing a chamber form a 2-DOF Helmholtz system with eigenmodes at frequencies that are not the simple geometric mean of the single-port frequencies. The coupling strength depends on port-area ratio and player hand impedance.

****Test:**** Build UDU-P2 (Medium), record audio at five hand conditions (mouth fully open / 25% / 50% / 75% / fully closed against a rigid hand seal, side hole open in each), spectrogram analysis of fundamental frequency vs hand condition. Compare to the placeholder coupled-mode eigenvalue model in `wolfram-starter.wl`.

****Mitigation:**** Empirically calibrate the coupling term in `wolfram-starter.wl` from the UDU-P2 measurements; record the calibrated coupling coefficient in the per-family corrections database via `scripts/record\_measurement.py`.

****Severity:**** Medium — affects predictive fidelity, not the build. Required before declaring the physics model "validated."

[Acoustic] Glaze fire shifts measured pitch — Medium

****Symptom:**** A bisque-tuned udu plays sharp/flat after glaze fire. Pitch shift typically 5–25 ¢ depending on glaze thickness and exterior coverage.

****Mechanism:**** Glaze adds wall mass (lowers shell-resonance damping slightly) and can subtly change the interior surface impedance at high frequencies. The Helmholtz fundamental itself is dominated by chamber volume × port area, but glaze film thickness can effectively reduce chamber volume by 1–3% on a 1/2" wall vessel.

****Test:**** Record mouth + side tones for UDU-P2 at three stages: greenware, bisque, glaze-fire. Log all three rows in `validation.csv` with environment (temp / RH).

****Mitigation:**** Cut ports slightly under-target at leather-hard; final-tune at bisque if needed; commit to glaze fire only when bisque tones are within ±25 ¢ of target. Glaze interior is forbidden (already called out in build).

****Severity:**** Medium — process-controllable, not a design defect.

Structural

[Structural] Wall thickness margin at side-hole edge — Medium

****Symptom:**** The side hole is cut at leather-hard and then enlarged at bisque for tuning. Each enlargement step removes wall material around the port. Repeated tuning cycles can leave the rim too thin (<1/8") for hand-slap durability.

****Mechanism:**** Hand slap on the port edge produces local stress ~10–20× the static body load. A 1/8" minimum wall is the rule for Cone-6 stoneware in domestic-use vessels; below that the slap-edge chips on first impact.

****Test:**** Slap-test a UDU-P0 port-tile with measured edge thicknesses

(0.30, 0.20, 0.15, 0.10 in). Document the thickness at which the rim chips after 100 slaps with a moderate hand.

****Mitigation:**** Limit total tuning enlargement to $\leq 30\%$ of original port area. Build a tuning trim *budget* into `validation.csv`: if predicted cents-error > 100 ¢, flag the body for rebuild rather than over-trim. Round all port edges with a damp finger / soft rib before bisque.

****Severity:**** Medium — manufacturing-controllable. Becomes High on UDU-XL where extreme cents-error correction is implied (see Acoustic risk #1).

[Structural] Greenware crack risk on long unsupported wall — Medium

****Symptom:**** UDU-XL's 14" diameter $\times$ 16" tall body has a 6"+ unsupported wall span between mouth and base after demold. Greenware shrinkage during drying produces hoop stress; cracks initiate at the side-hole notch and propagate vertically.

****Mechanism:**** Differential drying — interior surface dries more slowly than exterior — produces a stress gradient. Side-hole cut creates a stress concentrator. Stoneware tensile strength at 18–22% moisture content is ~ 50 PSI; predicted hoop stress at the crack location is ~ 30 – 40 PSI for the XL geometry.

****Test:**** First UDU-XL casting carries a strain gauge or dot-grid on the exterior wall; photograph at 24-hour intervals during drying; inspect side-hole crater for crack initiation. Repeat the same protocol on UDU-L as a sanity check.

****Mitigation:**** Slow-dry under plastic sheet for 5–7 days; cut side hole *after* initial drying, not at leather-hard. If cracks appear in the first XL, increase wall thickness to 0.40 in for that size only and re-emit XL drawings.

****Severity:**** Medium for UDU-S/M/L (geometry is well-supported), High for UDU-XL specifically — escalated.

Ergonomic

[Ergonomic] Side-hole height for seated player — Low

****Symptom:**** Player seated on a low stool reaches across an UDU-XL 14" diameter body to slap the side hole; reach distance ~12" puts the slap stroke off the player's natural arc.

****Mechanism:**** Standing or kneeling play accommodates the larger bodies; seated play (most common for solo recording) constrains the side-hole height to the 9th–10th-percentile player's natural arm-arc. For UDU-S/M the side hole at 5–6" height clears this; UDU-L/XL push the side hole to 7–8" which is the upper edge of the comfort band.

****Test:**** Mock up the UDU-XL profile in foam core; place on a 14" stand and have three players (5th-, 50th-, 95th-percentile arm reach) play the side hole for 60 seconds; rate comfort 1–5.

****Mitigation:**** Document recommended playing position per size in `assembly-manual.md`. UDU-XL is concert-bass with a stand; specify the stand height so the side hole sits at the seated player's elbow.

****Severity:**** Low — not a build defect; a documentation obligation.

[Ergonomic] Mouth rim sharpness on hand seal — Medium

****Symptom:**** Player seals the mouth with a flat hand strike. A sharp or undercut rim cuts the palm during repeated play; even a slightly acute angle causes redness on the heel of the palm after 10 minutes of continuous play.

****Mechanism:**** Hand seal needs full-area contact with no high points. The fired rim profile depends on the master geometry, demold cleanup, and any leather-hard finger-rolling. A clean radius $\geq 1/8"$ on the rim is the comfort minimum.

****Test:**** UDU-P0 cup tile carries three rim radius variants (1/16", 1/8", 1/4"). 5-minute continuous play at each; player rates comfort + check for skin redness.

****Mitigation:**** Specify rim radius $\geq 1/8$ " in the master CAD. Roll the rim with a damp finger / soft rib at leather-hard before bisque. Reject any body where the rim radius reads $< 1/16$ " after demold.

****Severity:**** Medium — affects long play sessions. Easy to fix at the master geometry, hard to fix after firing.

Supply

[Supply] Cone-6 casting slip shrinkage variance — Medium

****Symptom:**** Two batches of "Cone 6 stoneware casting slip" from the same supplier shrink 11.5% and 13.0% respectively. Master scale factor $1/(1 - \text{shrinkage})$ shifts by $\sim 1.5\%$ (from 1.130 to 1.149), enough to push a tuned body 30–40 ¢ off target.

****Mechanism:**** Shrinkage is a function of clay particle size, water content, organic content, and firing schedule. Manufacturers specify a **nominal** shrinkage; actual shrinkage is batch-dependent and the spec is rarely tighter than $\pm 1.5\%$.

****Test:**** Pour three shrinkage coupons ($4" \times 1" \times 0.25"$ bars) per new slip batch; bisque-fire all three; measure shrinkage; record in `validation.csv` shrinkage column.

****Mitigation:**** ****Re-derive**** the master scale factor from the **measured** batch shrinkage before printing each new master, not from the workbook's 12% assumption. The OpenSCAD master takes shrinkage as a parameter; this is already wired in, but the procedure must be called out in the assembly manual.

****Severity:**** Medium — process-controllable. Skipping the coupon test is the failure mode, not the slip itself.

[Supply] Bambu X1C build envelope vs UDU-L/XL master — High

****Symptom:**** Bambu X1C build volume is $\sim 256 \times 256 \times 256$ mm ($\sim 10"$ cube). UDU-L master half is 12" tall $\times$ 12" wide $\times$ 6" deep at 12% shrinkage

scaling — exceeds the print bed in two dimensions. UDU-XL is worse.

Mechanism: Master halves print as one-shot solids for surface finish; splitting them adds layer-line discontinuities that telegraph into the plaster mold and ultimately the cast surface.

Test: Open the UDU-L master.scad output in OpenSCAD, export STL sliced as a half, load in BambuStudio — confirm "out of build volume" warning before printing.

Mitigation: Three options:

1. **Multi-section masters** — split each L/XL master into 2–3 stackable sections, glue and surface-fill at the joint, sand smooth before plaster pour.
2. **Outsource print** — Shapeways, Craftcloud, or local Maker Nexus bay with a larger printer (Prusa XL, Voron 350+ build, 600 mm format).
3. **Defer L/XL until the print farm grows** — UDU-S and UDU-M fit the X1C; ship those first. (Already explicitly called out in the README "Status" table.)

Severity: **High** for UDU-L/XL — design constraint that must be resolved before Tier-3 CAD. Escalated to Tony.

Fit / Finish

[Fit/Finish] Glaze adhesion at port edge — Low

Symptom: Wax resist at the mouth and side-hole rim is meant to keep the rim glaze-free for hand contact. A thin or uneven wax application lets glaze creep onto the rim during dipping; the rim glaze then chips off in service or feels gritty under the hand.

Mechanism: Wax resist relies on the contact angle between liquid glaze and a hydrophobic film. Low-temperature wax (<140 °F drip point) re-wets if the bisque body is warm from a kiln just opened. Brush strokes leave thin spots.

Test: Wax-resist a bisque coupon along a pencil-marked line;

dip-glaze; bisque-fire; inspect line edge under raking light for glaze creep.

****Mitigation:**** Wax under cool conditions (≤ 80 °F). Two thin coats beat one thick coat. Inspect each rim before glaze fire; touch up wax gaps. Specify the wax type in `bom.csv` (currently "wax resist" — too generic).

****Severity:**** Low — easy to catch at QC, affects play comfort if missed.

[Fit/Finish] Master split-line transfer to cast body — Medium

****Symptom:**** The two-piece plaster mold has a parting seam where the mold halves meet. Casting transfers a fine ridge along this seam onto the greenware body. If the seam runs through the hand-contact zone, the player feels a subtle bump on each strike.

****Mechanism:**** The mold split is geometrically necessary for demold; no slip-cast body avoids it. The skill is to **route** the split line away from hand-contact surfaces.

****Test:**** Trace the proposed split line on the master CAD; overlay the hand-contact heat-map (mouth seal palm + side-hole slap fingers). Any overlap is a design issue; relocate the split line.

****Mitigation:**** Position split line on the front-back equator of the body, not on the side-hole face. Trim split-line ridge with a damp fettling tool at leather-hard; final smooth before bisque.

****Severity:**** Medium — design + process. Already addressed in the mold strategy of `design.md`; this risk register documents the **why**.

Cross-cutting / lineage

[Cultural] Public-facing copy must respect Igbo lineage — High (procedural)

****Symptom:**** Listing the udu in a recruiter-facing portfolio without the lineage attribution reads as appropriative or naïve. The udu is

not a generic ceramic bass pot.

Mechanism: Documentation is a public artifact even when the build is private. Recruiters, students, and other makers will read the README and the build-log site without context.

Test: Pre-publication review by a knowledgeable second reader (family member, cultural-sensitivity reader, or peer maker familiar with West African instrument lineage). Sign-off recorded in `notes/` before the build-log site goes live.

Mitigation: The README, `design.md`, and the build-log site each carry an explicit attribution paragraph. The license clause (CC-BY 4.0) is for the *engineering documentation*, not for the instrument concept itself. Already captured in the LICENSE file and design.md "Cultural And Product Note" — this risk register documents the *requirement*, not a defect.

Severity: High (procedural — block publication of build-log site until pre-pub review is signed).

Risk summary

ID	Category	Severity	Status
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ACO-01	Acoustic — family-extremes target mismatch	High	Escalated
ACO-02	Acoustic — coupled-mode unmodeled	Medium	Empirical loop
ACO-03	Acoustic — glaze-fire pitch shift	Medium	Process
STR-01	Structural — port-edge wall margin	Medium	Process
STR-02	Structural — XL greenware crack	Medium / High XL	Escalated for XL
ERG-01	Ergonomic — side-hole reach	Low	Documentation
ERG-02	Ergonomic — mouth rim sharpness	Medium	Master-CAD
SUP-01	Supply — slip shrinkage variance	Medium	Process
SUP-02	Supply — Bambu envelope vs L/XL	High	Escalated
FIT-01	Fit/Finish — glaze creep at port edge	Low	QC
FIT-02	Fit/Finish — split-line transfer	Medium	Master-CAD
CUL-01	Cultural — lineage attribution	High procedural	Pre-pub review

****High-severity, escalated to human:**** ACO-01 (family-extremes target mismatch), STR-02 (XL crack risk), SUP-02 (Bambu envelope), CUL-01 (pre-pub review).

These four block declaring the v4.1 udu packet "production-tagged" until Tony resolves them.